MW REALTY · TESTER HANDBOOK · PROJECT DOCUMENT P6

Environment & Setup

What a MW Realty test environment runs on, what must be running in the background, which settings and sandbox accounts it needs, and how the application behaves when a service is missing.

Scope: QA / UAT environments and developer machines. Covers supported browsers and devices, the server stack, required background processes, configuration keys (names only, never values), third-party sandbox accounts, time zone, currency and languages, and environment-specific behaviour testers should expect. Sources: `composer.json`, `package.json`, `vite.config.js`, `.env.example`, `bootstrap/app.php` and `config/*.php`. Secret values are never written in this document; the project lead shares them separately with the people who set up the environment.

1. Browsers and devices

The project does not define a browserslist or a Vite build target, so the build uses Vite 7's default target ("baseline widely available": roughly Chrome and Edge 107+, Firefox 104+, Safari 16+). Older browsers are not supported.

Platform	Browsers to test	Priority
Desktop (Windows / macOS)	Latest Chrome, latest Edge, latest Firefox, latest Safari (macOS)	Chrome every cycle; at least one other per cycle; all before release
iPhone / iPad	Safari on the latest two iOS versions	Every release
Android phone / tablet	Chrome on a recent Android version	Every release

Width (px)	Represents	Check
375	Phone	Menus open, tables scroll inside their box instead of overflowing, forms and pop-ups fit, the AI chat widget and compare bar do not cover buttons
768	Tablet	Sidebar and filter panels collapse correctly
1280	Laptop	Default layout for Admin and CRM
1920	Large desktop	No stretched or misaligned layouts

- Arabic must be checked: switching the website to Arabic turns the layout right-to-left. Check at least the home page, property search, property details (including the new insight tabs), compare, sign-up / log in, the customer profile and the AI chat (Arabic, Urdu and Persian bubbles are right-aligned).
- Voice input and read-aloud in the AI chat depend on browser support; a missing microphone button on some browsers is expected.
- The Admin (`/admin`) is a server-rendered Bootstrap 5 interface; the website and CRM are Vue single-page apps. Hard-refresh (or clear the cache) after a new build if a screen looks out of date.

2. Server stack

Layer	Technology	Version in the project
Language	PHP	^8.5 (Composer platform set to 8.5.0)
Framework	Laravel	^13.0
API authentication	Laravel Sanctum	^4.3 (Bearer tokens for the mobile apps)
Social sign-in	Laravel Socialite	^5.31 (Google, customers only)

Layer	Technology	Version in the project
Admin / CMS	`mightywarnerskochi/cms` package, Blade, Bootstrap 5, roles via spatie/laravel-permission	dev-main
Payments	Stripe PHP SDK	^21.3
Media	Cloudinary PHP (optional; local storage otherwise)	^3.1
Other backend	laravel-dompdf (invoices), Laravel Excel (imports / exports), libphonenumber-lite (phones), HTMLPurifier (safe rich text), bacon-qr-code (2FA QR)	see `composer.json`
Front end	Vue 3 and Vue Router 4 (website and CRM), axios, Tailwind CSS 4, Leaflet / MapLibre GL maps, GSAP, intl-tel-input	Vue ^3.5.42, Vue Router ^4.6.4
Build	Vite ^7.0.7 with laravel-vite-plugin ^2.0.0 and @vitejs/plugin-vue ^6.0.8; entry points `resources/css/app.css`, `resources/js/app.js`, `resources/js/crm/main.js`	Node version not pinned; Vite 7 needs Node 20.19+ or 22.12+
Database	MySQL (as in `.env.example`, database `mw_realty`); SQLite is the framework fallback	Chosen per environment
Queue	Laravel database queue (`QUEUE_CONNECTION=database`), plus a separate `imports` connection and queue for long imports	
Cache and sessions	Database (`CACHE_STORE=database`, `SESSION_DRIVER=database`, lifetime 120 minutes)	
Mail	`MAIL_MAILER=log` by default (emails written to the log); SMTP for a real test inbox	
API docs (developers)	Scribe ^5.11, static output in `public/docs/_scribe`	

3. Background processes

Process	How it runs	What it does	What breaks without it
Scheduler	Cron entry `* * * * * php artisan schedule:run` (from the project folder), or `php artisan schedule:work` on a developer machine	Every minute it also starts the queue workers below, and runs all timed jobs (table in 3.1)	Nothing timed happens: no queued emails, no imports, Premium never starts or ends, ads never settle, permits never expire, credits never expire, no renewal reminders
Default queue worker	Started by the scheduler every minute (`queue:work --stop-when-empty --tries=3 --max-time=50`); a permanent `php artisan queue:work` may run instead	Sends queued emails and notifications, Facebook lead imports, photo fingerprinting for Listing Review	Emails and some bell notifications never arrive; Listing Review finds no photo matches (unit-number matches still work); Facebook leads do not arrive
Imports queue worker	Started by the scheduler every minute (`queue:work imports --queue=imports --tries=1 --timeout=3600 --max-time=50`)	Lead imports, property imports (Excel / CSV), Property Finder syncs	Imports stay at "Waiting for the import worker"; after 15 minutes without progress they are marked failed
Public storage link	One-time `php artisan storage:link` (creates `public/storage`)	Serves uploaded files stored locally (photos, logos, documents, banners) when Cloudinary is not used	Uploaded images show as broken; downloads fail
Front-end build	`npm run build` after each deployment (or `npm run dev` locally)	Builds the website and CRM assets into `public/build`	Website and CRM show a blank page or old screens

> Testers: allow 1-2 minutes before reporting a missing email or a stuck import, and say in the report whether the scheduler and workers are known to be running on the environment.

3.1 Scheduled jobs

Job	When	Visible effect
`queue:work` (default and imports)	Every minute	Emails, notifications, imports processed

Job	When	Visible effect
`properties:expire-permits`	Daily 00:05 Dubai time	Listings with an expired permit go off the website; reminder 7 days before
`properties:expire-featured`	Every 15 minutes	Scheduled Premium starts; ended Premium is removed
`ads:settle`	Every 15 minutes	Unreviewed client ads past their end are closed and refunded
Newsletter campaign sender	Every 15 minutes	Due newsletter content is sent
`credits:expire`	Hourly	"Expired credits" lines in credit history
`currencies:sync-rates`	Hourly, only when `CURRENCY_RATES_AUTO_SYNC=true`	Currency rates updated from the feed
`visitors:prune`	Daily 03:30	Anonymous visitor activity older than 30 days deleted
`market-demand:aggregate`	Hourly at :10	Website activity rolled into anonymous demand counts (Buyer Demand, CRM Market Demand)
`facebook:check-page-tokens`	Daily 08:00	Broken Facebook Pages flagged; reconnect email
`portal:remind-subscription-renewals`	Daily 09:00 Dubai time	Email and bell the day before an auto-renewal
`portal:notify-expiring-documents`	Daily	Email when a passport, trade licence or TRN expiry date has passed
`landing-pages:cleanup-temp-images`	Daily	Removes unused temporary landing-page images

4. Configuration a QA environment needs

Values live in the environment's `.env` file, which is never committed and never shared in documents. The table lists key names only.

4.1 Core application

Key	Purpose	Required
`APP_NAME`, `APP_ENV`, `APP_KEY`, `APP_DEBUG`, `APP_URL`	Name, environment type, encryption key, error detail, base URL used in links and emails. Use `APP_ENV` other than `production` on QA and `APP_DEBUG=false` for realistic error pages.	Yes
`APP_LOCALE`, `APP_FALLBACK_LOCALE`	Default language (`en`)	Yes
`DB_CONNECTION`, `DB_HOST`, `DB_PORT`, `DB_DATABASE`, `DB_USERNAME`, `DB_PASSWORD`	Database (MySQL)	Yes
`SESSION_DRIVER`, `SESSION_LIFETIME`, `SESSION_DOMAIN`	Sessions (database, 120 minutes)	Yes
`QUEUE_CONNECTION`	Queue driver (database)	Yes
`CACHE_STORE`	Cache (database)	Yes
`FILESYSTEM_DISK`	Local file storage disk	Yes
`MAIL_MAILER`, `MAIL_HOST`, `MAIL_PORT`, `MAIL_USERNAME`, `MAIL_PASSWORD`, `MAIL_SCHEME`, `MAIL_FROM_ADDRESS`, `MAIL_FROM_NAME`	Outgoing email. `log` writes emails to `storage/logs/laravel.log`; SMTP to a test inbox is recommended for QA	Yes
`CMS_ADMIN_EMAIL`, `CMS_ADMIN_NAME`	Admin notification address and name used by the CMS	Recommended
`LOG_CHANNEL`, `LOG_LEVEL`	Logging	Yes

4.2 Security, sign-in and approval rules

Key	Purpose	Required
`OTP_ENABLED`, `OTP_STATIC_CODE`	`true` emails real sign-up / email-change codes; `false` sends none and accepts the fixed test code (also accepted at the new-device step outside production)	Optional (default true)
`TWO_FACTOR_REQUIRED` (older name `TWO_FACTOR_REQUIRED_FOR_AGENCIES`)	`true` (default): every agency and agent must set up 2FA, no Skip. `false`: optional unless the agency requires it	Optional
`RECAPTCHA_SITE_KEY`, `RECAPTCHA_SECRET_KEY`, `RECAPTCHA_MIN_SCORE`	Google reCAPTCHA v3 on public forms; blank = forms work unprotected	Optional
`LISTING_SUPERADMIN_APPROVAL`	`true`: every listing needs Super Admin approval on Listing Permits. `false` (default): a permit verified online goes live by itself; DTCM and None always need approval	Optional
`LISTING_REVIEW_MINUTES_PER_CHECK`	Minutes counted per screened listing in the "Time saved" measure (default 10)	Optional
`HOUSE_LICENSE_NAME`, `HOUSE_ORN`, `HOUSE_ADREC_LICENSE`	MW Realty's own licence details shown on house listings	Optional

4.3 Third-party services (use sandbox / test accounts only)

Service	Keys	Sandbox to request	If not configured
Stripe (plans, credit top-ups)	`STRIPE_KEY`, `STRIPE_SECRET`, `STRIPE_WEBHOOK_SECRET`, `STRIPE_CURRENCY` (aed)	Stripe account in test mode with test API keys and a webhook to `/api/stripe/webhook`	Choosing a paid plan becomes an upgrade request approved in Admin; card top-ups unavailable
Mail	`MAIL_*` keys above	Mailpit, Mailtrap or a shared test mailbox	`log` mailer: emails only in the application log
Google sign-in (customers)	`GOOGLE_CLIENT_ID`, `GOOGLE_CLIENT_SECRET`, `GOOGLE_REDIRECT_URI` (`https://mw.realty.mightywarnersdemo.com/customer/auth/google/callback`)	Google Cloud OAuth client for the QA domain	"Continue with Google" button hidden
Google Translate (auto-translate of bios and listings)	`GOOGLE_TRANSLATE_API_KEY`, `TRANSLATE_DRIVER` (google or mymemory), `MYMEMORY_EMAIL`	Google Cloud project with Cloud Translation API, or MyMemory for testing (small daily quota)	Translate options disabled ("Auto-translate isn't set up on this site yet")
AI chat, compare summary, Write with AI, import AI descriptions	`CHATBOT_PROVIDER` (groq or gemini), `GROQ_API_KEY`, `GROQ_MODEL`, `GEMINI_API_KEY`, `GEMINI_MODEL`, `CHATBOT_NAME` (default Noor)	A Groq or Google AI Studio key for the QA project	Fixed-text fallbacks (section 6)
Facebook Lead Ads	`FACEBOOK_APP_ID`, `FACEBOOK_APP_SECRET`, `FACEBOOK_WEBHOOK_VERIFY_TOKEN`, `FACEBOOK_GRAPH_VERSION`	A Meta developer app in development mode with test Pages; OAuth redirect and webhook URLs for the QA domain	Facebook connect fails; use the Facebook CSV import instead

Service	Keys	Sandbox to request	If not configured
Property Finder	`PROPERTY_FINDER_API_URL` (each account enters its own API key and secret in the CRM)	Test Property Finder Enterprise credentials from the project lead	Connection test fails; no sync
DLD Trakheesi (Dubai permits)	`DLD_PERMIT_API_ENABLED`, `DLD_PERMIT_API_URL`, `DLD_PERMIT_API_CLIENT_ID`, `DLD_PERMIT_API_CLIENT_SECRET`, `DLD_PERMIT_API_TIMEOUT`	DLD sandbox access (not yet available; default disabled)	Validate button hidden; RERA listings wait for Super Admin approval
ADREC (Abu Dhabi and Al Ain permits)	`ADREC_PERMIT_API_ENABLED`, `ADREC_PERMIT_API_URL`, `ADREC_PERMIT_API_CLIENT_ID`, `ADREC_PERMIT_API_CLIENT_SECRET`, `ADREC_PERMIT_API_TIMEOUT`	ADREC sandbox access (default disabled)	Same as DLD for ADREC listings
Cloudinary (media)	`CLOUDINARY_CLOUD_NAME`, `CLOUDINARY_API_KEY`, `CLOUDINARY_API_SECRET`, `CLOUDINARY_FOLDER`	A separate Cloudinary folder or account for QA	Uploads stored locally (needs `storage:link`)
Currency rates feed	`CURRENCY_RATES_AUTO_SYNC`, `CURRENCY_RATES_URL`	None (public feed)	Off by default: rates are whatever Admin typed
Maps	`MAP_STYLE_URL`, `MAP_ATTRIBUTION`, `MAP_FALLBACK_TILE_URL`	None (OpenFreeMap / OpenStreetMap defaults)	Defaults used
Market insights	`PROPERTY_INSIGHTS_MIN_SAMPLE`	None	Default 3 listings or deals per figure
AWS, Redis, Postmark, Resend, Slack	Standard Laravel keys	Not used by default	Not needed

> Never point a QA environment at production Stripe keys, a production Facebook app, real DLD / ADREC credentials or the production database or mailbox.

5. Time zone, currency and languages

- Application time zone: UTC (`config/app.php`). Business schedules that matter to users run in Dubai time (Asia/Dubai, UTC+4): permit expiry at 00:05, renewal reminders at 09:00, and ad bookings, which start on the hour in Dubai time. Record times in bug reports in UAE time.
- Currency: every price is stored in AED. The website currency switcher converts with rates kept in Admin > General Settings > Currencies (units per 1 AED); Stripe charges in AED. Compare adds a 1.5% conversion margin row for non-AED currencies.
- Website languages: English and Arabic text files exist (`resources/lang/cms-static/en.json`, `ar.json`); Admin enables languages and edits translations. Arabic switches the layout to right-to-left.
- AI chat languages: English, Arabic, Hindi, Urdu, Persian, Russian, Simplified Chinese, French, German and Spanish, independent of the website language.

6. Environment-specific behaviour (expected, not bugs)

Situation	Behaviour
No AI provider key	AI chat answers "The AI assistant isn't available yet - please use the enquiry form on any listing instead."; compare shows the rule-based "Quick summary"; Write with AI shows "AI writing isn't set up yet ..."; import's AI option disabled

Situation	Behaviour
AI provider error or rate limit	Chat error line "Sorry, I'm having trouble answering right now..."; compare falls back to the Quick summary
DLD / ADREC check disabled	No Validate button; RERA and ADREC listings wait on Listing Permits for Super Admin approval; Listing Review never raises "Ad differs from the permit record"
`LISTING_SUPERADMIN_APPROVAL=true`	Every listing waits for approval, even with a verified permit
Stripe not configured	Plan purchase becomes an upgrade request; credit top-up unavailable
Credits switched off in Admin	Buying refused; credits already held can still be spent
`OTP_ENABLED=false`	No code emails at sign-up; the fixed test code is accepted
`MAIL_MAILER=log`	No emails in any inbox; read them from the application log
Scheduler or workers stopped	Emails, imports, Premium start / end, ad settlement and expiries do not happen
No `storage:link` and no Cloudinary	Uploaded images broken
Few listings in a community	Market Insights shows "Not enough data yet"; Listing Review price checks need at least 5 similar listings
Currency auto-sync off (default)	Rates change only when Admin edits them
reCAPTCHA keys blank	Forms work without bot protection
reCAPTCHA keys set but PHP has no CA certificate file (common on local XAMPP)	Every protected form, including the /compare unlock, fails with "Verification failed — please try again."; the log shows "cURL error 60". Fix: set `curl.cainfo` and `openssl.cafile` in php.ini to a CA bundle (XAMPP: `apache/bin/curl-ca-bundle.crt`) and restart PHP
reCAPTCHA key's allowed domains do not include the test host	"Verification failed — please try again." on every protected form; add the host (e.g. `127.0.0.1`, `localhost`) in the reCAPTCHA admin console
Several testers on one IP address	Shared rate limits ("too many requests") are reached sooner
Cached configuration on the server	A changed `.env` value has no effect until a developer clears and rebuilds the config cache